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Mycosynthesis of silver nanoparticles using *Pleurotus florida*: characterization and evaluation of antimicrobial efficacy

Swati Jha* & Arun Kumar

University Department of Botany, Ranchi University, Ranchi, Jharkhand, India

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Abstract- In the present era of rapidly growing technology, nanoparticle section has managed to draw a considerable attention. The present study shows the eco-friendly synthesis of silver nanoparticles through aqueous extract of *Pleurotus florida*. The colour change to dark brown, after adding silver nitrate, marked the formation of nanoparticles. After the initial screening, further characterization was carried on through UV-Visible spectroscopy, Fourier Transform Infrared Spectroscopy (FTIR) and Scanning Electron Microscopy (SEM). UV analysis depicted absorption peak range of 410-418 nm. The irregular or quasi spherical shape of the particles and that too in nanometre range was confirmed by SEM analysis. Various functional groups were confirmed through FTIR analysis that reduced and further stabilised the nanoparticles. Now, as an application of synthesized nanoparticles, this study also demonstrated the antimicrobial efficacy against selected micro-organisms. The findings of the present study highlight the potential role of *Pleurotus florida* to work as an effective biological source for the green synthesis of silver nanoparticles with promising effects.

Keywords: *Pleurotus florida,* Silver nanoparticles, UV-Vis spectroscopy, SEM, FTIR, Green synthesis, Antimicrobial efficacy.

INTRODUCTION

Nanotechnology has become one of the very important aspects of consideration for the researchers. It is becoming one of the fastest emerging areas of modern scientific research owing to its diverse applications in the fields like medicine, electronics, environment and agriculture. Different metals are used for the synthesis of nanoparticles, but amongst all, silver nanomaterials are widely prepared and taken into use. The reason lies in the chemistry of silver metal. Silver is a d- block transition metal belonging to 5th period and group 11. It is a soft, whitish-grey, lustrous metal with atomic number 47 and weighing 107.87 amu. It is found in Earth's crust in free elemental form as Ag, or in alloys with gold and other metals or in minerals such as "Argentite" and "Chlorargyrite". Besides exhibiting excellent thermal and electrical conductivity, silver is used in making jewellery owing to its lustre and non- reactivity which is further owed due to its fully filled 4d shell which is not effective in shielding the very last 5s electron from the forces of nuclear attraction. Silver has been long recognized as a strong antimicrobial agent due to the unique characters of Ag^{+}. Silver readily form complexes with certain groups like thiols in proteins, sulphur and phosphorus containing molecules in the cells. These all may lead to enzyme inactivation, interference in important mechanism like DNA replication and cell division and disturbances in microbial respiration. So, silver exhibit a broad spectrum antimicrobial

*Corresponding author :
Phone : 7909014002
E-mail : 13swatishashwati@gmail.com

effects even at a very low concentration. These altogether makes silver, the best candidate for nanoparticle synthesis.[1,2] Silver nanoparticles are widely explored for their applications medical devices, drug delivery systems and antimicrobial coatings. Conventional chemical method uses chemicals as reducing agents, while the eco-friendly green synthesis employs the use of phenolics, polysaccharides and other biomolecules as reducing and stabilizing agents for the nanoparticle synthesis. Chemical methods of silver nanoparticle synthesis may prove hazardous as it involves high energy, elevated temperatures and harmful chemicals that may pose a potential risk to both humans and environment. So, nowadays, attentions have been diverted towards the green synthesis of nanoparticles for a clean environment and promote sustainable development.[3,4]

Fungi are rich in metabolic diversity and possess excellent ability to secrete bioactive compounds. So, fungal systems are widely explored for the synthesis of nanoparticles. Again, what makes them a suitable candidate: It is the presence of large quantities of extracellular enzymes, proteins and secondary metabolites that can effectively reduce metal ions and stabilize the resulting nanoparticles. There are several findings that show that fungal extracts are full of certain bioactive compounds that can act as both capping as well as stabilizing agents during nanoparticle synthesis.[5] While talking about fungi, mushrooms has to be the centre of attraction. Edible mushrooms form the richest sink of proteins, especially for vegetarians.

Now, while exploring the other facets of mushrooms, we come across their potential as biological resource fort the nanoparticle synthesis due to their richness in proteins, phenols, polysaccharides and important enzymes. In the arena of mushrooms, *Pleurotus* is the most widely cultivated and consumed mushroom, attributed to its rich medicinal properties and easy cultivation.[6,7] Among the various species of *Pleurotus, P.florida* is found to be rich in several bioactive molecules and metabolites that impart significant anti-oxidant and anti-microbial properties to them.[8,9] Now, considering the growing interest in the nanotechnological field, and particularly in the maintaining the eco-friendliness in the process, the present study is also designed in the same manner of green synthesis of silver nanoparticles while using an important edible fungus. Additionally, they were characterized using UV-Vis spectroscopy, SEM analysis, and FTIR, followed by antimicrobial evaluation.

MATERIALS & METHODS

The spawn of *Pleurotus florida* was obtained from "Birsa Agriculture University" Kanke, Ranchi, Jharkhand.

The cultivation of *Pleurotus florida* was carried on at the mushroom unit in BAU, Ranchi. Fruiting bodies were collected, weighed, powderd and stored. (Fig.1).



Fig. 1: Showing fruiting bodies of *Pleurotus florida*

Classification:

Kingdom	-	Fungi
Phylum	-	Basidiomycota
Class	-	Agaricomycetes
Order	-	Agaricales
Family	-	Pleurotaceae
Genus	-	*Pleurotus*
Species	-	*florida*

Formation of mushroom powder:

Freshly harvested mushrooms were thoroughly washed to remove impurities, cut into thin slices thereafter dried in shade for 3-4 weeks. Then, they were crushed and powered in a mortar and pestle.

Preparation of aqueous extract of mushroom:

Dried mushroom powder was used for the extraction purpose. The powder was weighed and placed in the thimble lined with cheese cloth and the extraction was performed using a Soxhlet apparatus with distilled water as the extraction solvent. The solvent was heated in a round-bottom flask attached to the Soxhlet extractor and condenser, allowing continuous extraction for approximately 6h. Using a rotatory evaporator, the obtained extract was concentrated and dried to obtain the crude extract.

Synthesis of Silver nanoparticles:

The silver nanoparticle experiment was carried at the Cytogene Research Laboratory, Lucknow, UP, India. The synthesis was done based on previous studies with slight modification.[10,11]

For the synthesis purpose, 400 mg of mushroom extract was dissolved in 2 ml distilled water which was further mixed with 180 ml of 5 mM silver nitrate solution along with pH adjustment to approx. 9 using, 0.1M sodium hydroxide solution. The solution maintenance was done at 60°C followed by magnetic stir at 150 rpm for 2 hours.

Characterization of silver nanoparticles:

UV-Vis Spectroscopy:[11]

The confirmation of silver nanoparticle formation was done using UV-Vis spectrophotometry. The wavelength of the recorded absorbance spectrum was in the range of 300-600 nm.

Scanning Electron microscopy (SEM) analysis:[2,12]

The dried nanoparticle samples were mounted on the suitable stubs and examined under SEM. The morphology and surface characteristics of the synthesized nanoparticles were observed through scanning electron microscopy.

Fourier Transform Infrared (FTIR):[11]

FTIR analysis was performed to find the groups attached to the synthesized silver nanoparticles.

Antimicrobial Activity:[13-16]

The antimicrobial activity of the synthesized silver nanoparticles obtained from *Pleurotus florida* extract though the well diffusion method. The experiment was done against *Staphylococcus aureus* (MTCC 3160), *Escherichia coli* (MTCC 452), *Salmonella typhi* (MTCC 733) and *Candida albicans* (MTCC 183). Different media was used for both bacteria and fungus. Muller-Hinton Agar was used for the bacteria and Sabouraud Dextrose Agar for the fungal strain. The media prepared as sterilised by autoclaving at 121°C and 15 psi pressure for 15 min. Further they were poured in the petri plates and allowed to solidify. Then, the microbial cultures were evenly spread over the plates and allowed to multiply. A 6 mm diameter wells were prepared with the help of a cork borer and three different concentrations of the silver nanoparticles suspension, that is, 25 µg/mL, 50 µg/mL and 100 µg/mL, were filled in the wells created. Ciprofloxacin (100 ppm) and Itraconazole (300 ppm) were used as positive controls for bacteria and fungi respectively, while distilled water was kept as the negative control. The plates were incubated at 37°C for 24 h, after which the zones of inhibition were measured in millimetres (mm).

RESULT & DISCUSSION

Formation of silver nanoparticles:

The initial indication of silver nanoparticles formation by using the aqueous extract of *Pleurotus florida* was confirmed by visible colour change of the reaction mixture. Upon the addition of silver nitrate solution, the colour of the reaction mixture gradually changed from yellow to dark brown, indicating the reduction of silver ions (Ag^+) to silver nanoparticles (AgNPs) (Fig. 2)

UV-Vis spectroscopic analysis:

The formation of silver nanoparticles was further confirmed by UV–Visible spectroscopic analysis. The nanoparticles exhibited a characteristic surface plasmon resonance (SPR) peak in the range of 410–418 nm, which is typical for mushrooms. The maximum absorbance was observed at 418 nm, confirming the successful synthesis of AgNPs from *Pleurotus florida* extract (Fig. 3)

SEM Analysis:

Scanning Electron Microscopy (SEM) method was employed to study the surface morphology of the synthesized silver nanoparticles. The SEM micrograph revealed that the nanoparticles were predominantly irregular to quasi-spherical in shape and were present in the nanometre size range. The particles appeared to form small aggregates, which are commonly observed in biologically synthesized nanoparticles due to the presence of biomolecules acting as capping agents. The SEM image confirmed the successful formation of silver nanoparticles synthesized using *Pleurotus florida* extract (Fig. 4).

FTIR Analysis:

Fourier Transform Infrared Spectroscopy (FTIR) was performed to find the groups attached to the nanoparticles synthesized. These groups are may be responsible for the reduction and stabilization of the nanoparticles. The spectrum showed prominent absorption peaks at 3784.45, 3700.35 and 3235.52 cm^{-1}, corresponding to O–H and N–H stretching vibrations that indicated the presence of hydroxyl and amine groups associated with phenolic compounds and proteins. Further peaks observed at 1605.08 cm^{-1} was attributed to aromatic C=C or amide I vibrations. The proteinaceous biomolecules were confirmed by absorption band at 1521.03 cm^{-1}

attributed to amide II vibrations. Further peaks in the region of 1337.95–1024.94 cm^{-1}, corresponding to C–N and C–O stretching vibrations, indicated the presence of alcohols, phenols and polysaccharides. Aromatic ring vibrations were confirmed by observation of absorption bands observed below 900 cm^{-1} (Fig. 5).

Antimicrobial Activity:

The silver nanoparticle extract of *Pleurotus florida* were tested against four different micro-organisms namely, *Staphylococcus aureus* (MTCC 3160), *Escherichia coli* (MTCC 452), *Salmonella typhi* (MTCC 733) and *Candida albicans* (MTCC 183). Maximum zone of inhibition was observed against *Salmonella typhi* measuring 16.50 ± 2.17 mm at 100 µL, indicating the highest antibacterial activity. Secondly, it was observed in *Staphylococcus aureus* (15.33 ± 0.58 mm) followed by *Escherichia coli* (14.66 ± 0.28 mm) at the same concentration. However, inhibition zones were smaller compared to those produced by positive control ciprofloxacin which were 26.8 ± 1.89 mm for *S. typhi*, 30.00 ± 0.00 mm for *S. aureus*, 21.33 ± 2.30 mm for *E. coli* (Table 1). Antifungal activity was evaluated against *Candida albicans* which showed a zone of inhibition of 14.50 ± 0.86 mm at 100 µL concentration. In comparison, itraconazole produced a larger inhibition zone of 25.00 ± 0.00 mm. No zones were observed in negative control (Table 1-2, Fig. 6).



Fig. 2: Visual colour change during the synthesis of silver nanoparticles using *Pleurotus florida* extract. (A) Reaction mixture before nanoparticle formation. (B) Formation of silver nanoparticles indicated by dark brown colour.

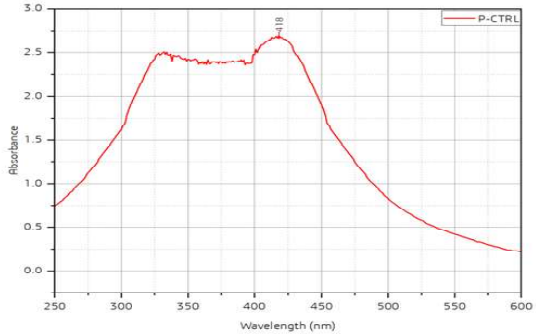


Fig. 3: UV-Vis absorption spectrum of silver nanoparticles synthesized from *Pleurotus florida* extract

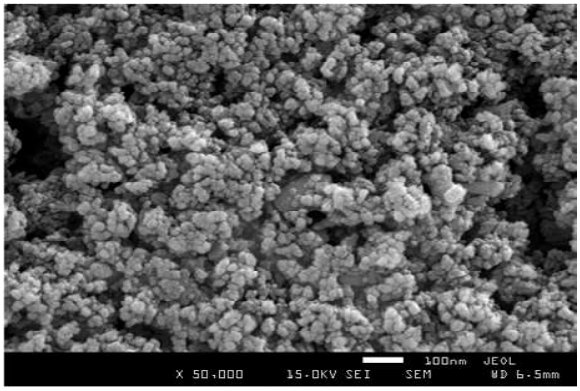


Fig. 4: SEM micrograph of silver nanoparticles synthesized from *Pleurotus florida* extract

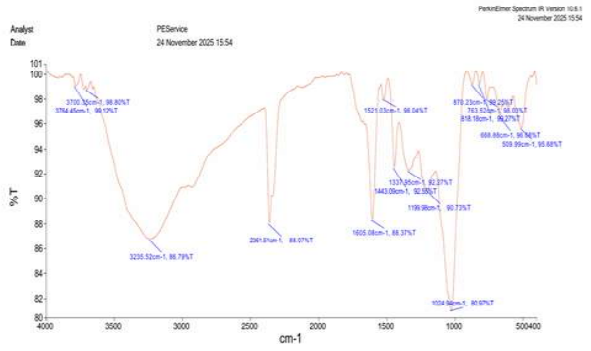


Fig. 5: FTIR spectrum of silver nanoparticles synthesized from *Pleurotus florida* extract



Fig. 6: Antimicrobial activity of *Pleurotus florida* against selected microbes showing zones of inhibition.

Table 1: Antibacterial activity of biosynthesized silver nanoparticles derived from *Pleurotus florida* extract

Code No.	Bacterial Strains	Zone of inhibition in mm				
		25µl	50µl	100µl	T(PC)	E(NC)
MTCC3160	*S. aureus*	10.67±0.58	12.67±0.58	15.33±0.58	30.00±0.00	nil
MTCC452	*E. coli*	13.16±0.28	14.00±0.00	14.66±0.28	21.33±2.30	nil
MTCC733	*S. typhi*	15.60±2.08	15.60±2.08	16.50±2.17	26.8±1.89	nil

Table 2: Antifungal activity of biosynthesized silver nanoparticles derived from *Pleurotus florida* extract

Code No.	Fungal strain	Zone of inhibition in mm				
		25µl	50µl	100µl	T(PC)	E(NC)
MTCC183	*C. albicans*	14.00±1.00	14.30±1.15	14.50±0.86	25.00±0.00	nil

Nanoparticle formation was initially screened out by visual observation of colour change yellowish brown to dark brown. Similar observation is reported in previous studies.[4,11,17,18] This colour change is attributed to reduction of silver ions (Ag^{+}) to metallic silver (Ag^{o}). The UV-Vis spectroscopic absorption peak of the range 410-418 nm is in accordance with some earlier findings.[2,11] SEM analysis revealed the quasi spherical shape of nanoparticles and small aggregates that is often visible in biological synthesis, present in nanometer size range and the morphology observed is also in agreement with earlier studies.[11] FTIR analysis revealed the presence of many functional groups that acted as good stabilizing and capping agents in these nanoparticles prepared from *Pleurotus florida,* which is further in agreement with earlier findings.[3,4,19] The silver nanoparticles of *Pleurotus florida* exhibited moderate antimicrobial activity against the tested microorganisms. The highest inhibition was observed against *Salmonella typhi* (16.50 ± 2.17 mm), followed by *Staphylococcus aureus* (15.33 ± 0.58 mm) and *Escherichia coli* (14.66 ± 0.28 mm). These inhibition zones fall within the range commonly reported for biologically synthesized silver nanoparticles, where inhibition zones of approximately 10–20 mm have been observed against various bacterial pathogens.[20-22] The present study is also in agreement with a previous report where they demonstrated a significant antibacterial activity of *Pleurotus florida*-derived silver nanoparticles against *Staphylococcus aureus* and *Escherichia coli.*[22] The observed antibacterial activity may also be influenced by structural differences in the cell walls of micro-organisms. The difference in morphology and composition of the cell wall of gram-negative bacteria, imparts a greater susceptibility to silver nanoparticles, as their thin peptidoglycan layer facilitates easier penetration of nanoparticles into the cell.[21]

CONCLUSION

The present study demonstrated the potential use of *Pleurotus florida* in synthesizing silver nanoparticles and also their successful application in the field of antimicrobial evaluation. The formation of nanoparticles were confirmed by UV-Vis spectroscopy, SEM and FTIR analyses, indicating the presence of bioactive compounds that played an important role as reducing and stabilizing agents. The synthesized nanoparticles exhibited significant properties of antibacterial and antifungal against potential microbes. Although the biosynthesis of silver nanoparticles using *Pleurotus florida* has been reported in previous studies, reports utilizing mushrooms cultivated in the Jharkhand region are limited. This approach also highlights the potentials of locally cultivated mushrooms in silver nanoparticle synthesis and this way chemical synthesis can be discouraged and sustainability can be maintained.

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